

KESARI DASARADHA RAMI REDDY

Python Backend Developer | FastAPI · PostgreSQL · REST APIs

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PROFESSIONAL SUMMARY

Results-driven **Python Backend Developer** with nearly **3 years of experience** designing and building **clean, scalable, and production-grade backend systems** using Python, FastAPI, SQLAlchemy, and PostgreSQL. Strong foundation in **OOP, design patterns, and data structures**, with hands-on expertise in **RESTful API development, pseudo-microservices architecture**, and third-party system integrations. Demonstrated ability to optimize database performance, enforce code quality standards, and implement **security best practices** across the full development lifecycle. Comfortable working with **cloud platforms (GCP)**, CI/CD pipelines, Docker, and Linux environments. A collaborative team player with a proven record of delivering reliable backend solutions aligned with product and QA timelines.

PROFESSIONAL EXPERIENCE

Python Backend Developer

KaryaHub Solutions

August 2023 – Present

Hyderabad, India

- Built and maintained **100+ clean, testable, and well-documented RESTful APIs** using **FastAPI** with sub-200ms latency across HRMS, ERP, and ticketing platforms, integrating seamlessly with frontend and third-party systems.
- Refactored a tightly coupled monolithic codebase into **Dockerized modular services** following **microservices architecture principles** behind an **NGINX reverse proxy**, cutting deployment time from ~2 hours to ~15 minutes (**87% improvement**) and boosting horizontal scalability by 40–50%.
- Applied **OOP principles and established design patterns** (Repository, Factory, Dependency Injection) throughout the codebase to maximize reusability, testability, and maintainability.
- Engineered a secure authentication layer using **OAuth2, JWT (RS256), and fine-grained RBAC**, combined with **AES-256 encryption** for sensitive data, aligned with GDPR and PCI-DSS practices.
- Designed and optimized **PostgreSQL schemas, indexes, and complex queries**; executed zero-downtime migrations using **Alembic**, and managed Linux-based server environments for staging and production.
- Built and maintained **CI/CD pipelines** using GitHub Actions, integrating **SonarQube, Snyk, Trivy, Bandit, and Dependabot** — detecting 30% more critical issues pre-release and enabling a **40% faster release cadence** in collaboration with QA and DevOps teams.
- Integrated **HashiCorp Vault** for centralized secrets management, eliminating hardcoded credentials and improving overall security posture by approximately 45%.
- Introduced **Redis-based caching and API rate limiting** to protect endpoints from abuse, reducing unnecessary database load and improving response times.
- Provisioned and managed **GCP infrastructure** (Compute Engine, Cloud SQL, VPC, IAM, Load Balancers) using **Terraform (IaC)**, with working familiarity in AWS services (EC2, S3).
- Authored comprehensive **API documentation via Swagger/OpenAPI and MkDocs**, writing reusable and well-documented code that improved cross-team onboarding efficiency by 35%.
- Participated in **code reviews** and enforced coding standards (Black, isort, Flake8, Pylint), troubleshooted complex production issues, and mentored 3 junior developers on async Python patterns, Git workflows, and API security — reducing onboarding time by 35%.

PROJECTS

Neuropi — School Management ERP

Python · FastAPI · Async SQLAlchemy · PostgreSQL · Redis · Docker · Pydantic v2 · Alembic

- Built **100+ clean and testable async REST APIs** across admissions, attendance, library, and document workflows, serving **5,000+ students** with consistent reliability.
- Migrated a synchronous legacy backend to an **async-first architecture** using AsyncIO, resolving event-loop bottlenecks and improving deployment stability by **50%**.
- Implemented **fine-grained RBAC** with strict request/response validation via Pydantic v2, reducing data inconsistencies by **95%**.

- Applied Redis caching and rate limiting to optimize performance and protect authentication endpoints from brute-force attacks.
- Designed environment-aware file storage with metadata persisted in PostgreSQL, supporting both local and cloud deployments.

• SHOU — HRMS and Animation Production Pipeline Tracker

Python · FastAPI · PostgreSQL · Docker · NGINX · Alembic · SonarQube · Snyk

- Developed a scalable **HRMS backend** with payroll, attendance, leave, and reporting modules processing **5,000+ monthly payroll records with 100% accuracy**.
- Designed an animation production pipeline tracker to manage project phases, asset assignments, and milestone workflows for creative teams.
- Refactored a legacy system into a **Docker-based modular microservices architecture**, significantly improving deployment reliability.
- Strengthened security with **JWT lifecycle management and RBAC**, reducing unauthorized access attempts by **99%**.
- Integrated automated security scanning (SonarQube, Snyk) and enforced zero-downtime database migrations using Alembic.

• Events2Go — Event Discovery and Ticketing Platform

Python · FastAPI · Async SQLAlchemy · PostgreSQL · Redis · Docker · AWS S3 · Stripe

- Designed a high-concurrency **async backend** supporting **10,000+ concurrent users** with event discovery, venue management, and multi-stage booking workflows integrated with frontend teams.
- Built a **concurrency-safe seat reservation engine** using PostgreSQL transactions and row-level locking, eliminating double-booking under peak traffic.
- Implemented a **multi-tenant RBAC model** enforced via Pydantic v2 validation for platform admins, organizers, and end users.
- Integrated **Stripe payment processing** with encrypted workflows ensuring PCI-DSS-aligned secure transactions.
- Reduced database load by **35%** via async query optimization and Redis caching while maintaining sub-200ms API latency.
- Wrote **50+ async Pytest** test cases covering booking logic, payment edge cases, and concurrency scenarios.

TECHNICAL SKILLS

- **Languages:** Python (primary), SQL
- **Frameworks & Libraries:** FastAPI (primary, sync/async), SQLAlchemy ORM, Pydantic v2; familiar with Django and Flask
- **Backend Engineering:** RESTful API Design, Microservices Architecture, Async Programming (AsyncIO), OOP & Design Patterns, Request/Response Validation, Third-Party Integrations, Swagger/OpenAPI, MkDocs
- **Databases & Caching:** PostgreSQL (schema design, indexing, query optimization), Redis (caching, rate limiting), Alembic; familiar with MySQL and MongoDB
- **Authentication & Security:** OAuth2, JWT (HS256/RS256), RBAC, AES-256 Encryption (Fernet), HashiCorp Vault
- **DevOps & Tools:** Docker, GitHub Actions (CI/CD), NGINX, Git, Linux/Unix environments
- **Cloud Platforms:** Google Cloud Platform (Compute Engine, Cloud SQL, VPC, IAM, Load Balancers), AWS (S3, EC2, Lambda — familiar)
- **Testing & Code Quality:** Pytest (unit, integration, async), Postman, Black, isort, Flake8, Pylint, SonarQube, Snyk, Trivy, Bandit

EDUCATION

• Bachelor of Technology (B.Tech) — Electrical & Electronics Engineering

Vasireddy Venkatadri Institute of Technology (VVIT)

2018 – 2022

Guntur, India